INCH-POUND

MIL-PRF-32814

29 June 2026

PERFORMANCE SPECIFICATION

POWDER FEEDSTOCK, COPPER-NICKEL (70-30), FOR USE IN LASER POWDER BED FUSION, ADDITIVE MANUFACTURING

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers copper-nickel (CuNi) (70-30) pre-alloyed powder. This powder is intended for use as a feedstock in laser powder bed fusion (LPBF) additive manufacturing, but usage is not limited to such application.

1.2 Safety – hazardous materials. This specification does not address the hazards or risk involved with handling metallic powder. It is the sole responsibility of the user to ensure proper use and handling.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B215	-	Standard Practices for Sampling Metal Powders
ASTM E478	-	Standard Test Methods for Chemical Analysis of Copper Alloys
ASTM E1941	-	Standard Test Method for Determination of Carbon in Refractory and Reactive Metals and Their Alloys by Combustion Analysis
ASTM E2465	-	Standard Test Method for Analysis of Ni-Base Alloys by Wavelength Dispersive X-Ray Fluorescence Spectrometry
ASTM E2575	-	Standard Test Method for Determination of Oxygen in Copper and Copper Alloys by Inert Gas Fusion

(Copies of these publications are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS7025	-	Metal Powder Feedstock Size Classifications

(Copies of this publication are available online at www.sae.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical composition. Chemical composition of powder shall conform to the percentages by weight specified in table I. Oxygen, tellurium, thallium, and selenium shall be measured and reported for information.

TABLE I. Chemical composition (weight percent).[1/]

	Minimum	Maximum
Copper	Remainder	
Lead	---	0.01
Iron	0.75	1.5
Phosphorus	---	0.02
Nickel plus cobalt	28.0	32.0
Carbon	---	0.15
Manganese	---	1.5
Sulfur	---	0.02
Silicon	---	0.5
Niobium	0.65	1.5

NOTES:

1/ Oxygen, tellurium, thallium, and selenium shall be reported for information only.

3.2 Powder production. Unless otherwise specified (see 6.2), powder shall be produced by any method that ensures homogeneous alloy composition throughout each particle. Surface-coated powders, elemental powder blends, or any combination thereof are prohibited. If atomization is used, all atomization shall be performed using a fixed gas composition consisting primarily of argon, helium, or nitrogen. Water atomization is not permitted. The particle size distribution (PSD) shall be in accordance with the fine and coarse tolerances or limits for the metal powder feedstock size class in accordance with SAE AMS7025, as specified (see 6.2). For powder produced by melting:

a. All melting shall be performed under vacuum or a fixed inert gas composition.

b. Air melting is not permitted.

c. Purchased scrap shall not be used in the furnace charge unless it is traceable to individual heats with certified chemical analysis conforming to table I, and unless it is free from contaminants that are deleterious to powder or parts fabricated using the powder.

3.3 Powder lot identification. Each powder lot shall have a unique identification (see 6.4.2 for definition of powder lot).

3.4 Powder reuse. Powder reuse is permitted. Powder that is reused shall meet all requirements of this specification.

3.5 Powder blending. Blending multiple powder lots to create a new powder lot is permitted provided every powder lot to be blended conforms to this specification at the time of blending, and the combined blended lot also conforms. Blending of powder from nonconforming lots to make a conforming lot is prohibited. Additionally, combined lots shall be manufactured by the same vendor, shall be produced by the same powder production method, shall be of the same size class as defined by SAE AMS7025 (see 3.2), and shall have a unique powder lot identification (see 3.3).

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3.6 Detrimental material controls.

3.6.1 Powder handling controls. All powder shall be controlled to prevent contamination that is deleterious to powder or parts fabricated using the powder. This includes atomization, screening, blending, and storage in addition to other processes completed by suppliers performing work to this specification.

3.6.2 Powder quality. The powder shall be uniform in color and quality, dry, and free from foreign materials and from imperfections detrimental to producing additively manufactured parts.

3.6.3 Powder purity. No substance shall be added to the powder at any time to aid powder flowability, spreadability, or any other powder characteristic.

3.6.4 Additional detrimental material controls. When needed, additional detrimental material controls shall be as specified (see 6.2).

3.7 Additional powder characterization. Additional powder characterization shall be as specified (see 6.2).

3.8 Powder containers. The containers used to store powder shall be suitable to prevent contamination or degradation from sources including foreign object debris, other powders, undesired lots, moisture, or oxygen ingress. To protect the contents from contamination during shipment and under normal dry storage conditions, each container shall be clean and dry, and shall be sealed immediately after filling. Security seals used on the individual containers shall be so designed that they are destroyed when opening the container.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

4.2 Conformance inspection. Conformance inspection shall include the examinations and tests of table II.

TABLE II. Conformance inspections.

Examinations and tests	Requirement	Conformance inspection	Sampling requirements
Chemical composition	3.1	4.4 and 4.7	4.3
Powder production	3.2	4.7	---
Powder lot identification	3.3	4.7	---
Powder reuse	3.4	4.7	---
Powder blending	3.5	4.7	---
Powder handling controls	3.6.1	4.7	---
Powder quality	3.6.2	4.7	---
Powder purity	3.6.3	4.7	---
Additional detrimental material controls	3.6.3	4.7	---
Additional powder characterization	3.7	4.6 and 4.7	---
Powder containers	3.8	4.7	---

4.3 Sampling for chemical composition. Unless otherwise specified (see 6.2), powder samples shall be taken in accordance with ASTM B215. A minimum of one sample shall be taken from each powder lot for analysis. Unless otherwise specified (see 6.2), the frequency for sampling and recertification of reuse powder (see 6.4.3 for definition of reuse powder) shall be every 10 build cycles, at a minimum. The powder sample shall be tested in the condition in which it is intended to be reintroduced into the build cycle (e.g., post-sieve) (see 6.4.1 for definition of build cycle).

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4.4 Chemical composition analysis. The chemical composition of each lot shall be tested and analyzed by accepted analytical methods. In case of disagreements, the referee methods shall be ASTM E2575 for oxygen, ASTM E1941 for carbon, ASTM E2465 for niobium, and ASTM E478 for the remaining compositional requirements. The chemical composition of each lot shall conform to the composition shown in table I.

4.5 Retesting. If a conformance test fails to meet the specified requirements, disposition of the powder may be based on the results of testing three additional samples for each original nonconforming test result. To conform to this specification, all three retests must conform. Failure of any one retest powder sample to meet the specified requirements shall be cause for rejection. Results of all tests shall be reported. One-to-one replacements for testing malfunctions not conforming to the analysis method are permitted.

4.6 Additional powder characterization. Test methods for additional powder characterization shall be as specified (see 6.2 and 6.3).

4.7 Certificate of conformance. Unless otherwise specified (see 6.2), a certificate of conformance (COC) shall be provided for each lot of powder offered for acceptance. The COC shall include the following:

a. Statement that the powder lot has been manufactured, sampled, tested, and inspected in accordance with the requirements specified herein (see 3.2).

b. The specified PSD and tolerance limits by the purchaser and the reported PSD from the powder feedstock manufacturer (see 3.2).

c. The unique powder lot identification (see 3.3).

d. If powder has been reused, statement that powder has been reused, and sampling frequency for retesting (see 3.4 and 4.3).

e. If powder has been blended, statement that powder has been blended, and the COCs from the parent powder lots (see 3.5).

f. Statement that all powder handling controls have been followed to prevent contamination (see 3.6.1).

g. Statement that the powder is uniform in color and quality and dry (see 3.6.2).

h. Statement that the powder has been handled in a manner to prevent the introduction of foreign materials and imperfections (see 3.6.2).

i. Statement that no substances have been added to the powder (see 3.6.2).

j. When additional detrimental material controls are required (see 6.2), statement certifying that requirements have been met (see 3.6.4).

k. Statement certifying that requirements for powder containers have been met (see 3.8).

l. A complete record of chemical analysis (see 4.4) and any other powder characterization performed under the order or contract (see 6.2 and 6.3).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Powder produced in accordance with this specification is intended to be used as feedstock to produce material for use in applications specified in MIL-PRF-32815, "Copper-Nickel (70-30), Laser Powder Bed Fusion, Additively Manufactured."

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Acceptable powder production methods, if applicable (see 3.2).

c. Particle size distribution requirements, if applicable (see 3.2).

d. When needed, additional detrimental material control requirements (see 3.6.4).

e. Additional powder characterization requirements and test methods, if applicable (see 3.7, 4.6, and 6.3).

f. Additional sampling requirements, if applicable (see 4.3).

g. Powder reuse sampling frequency, if applicable (see 4.3).

h. When a certificate of conformance is not required (see 4.7).

i. Packaging requirements (see 5.1).

6.3 Additional powder characterization. The characteristics of powder used in the LPBF process are known to affect performance outcomes. Metrics such as flowability, particle size distribution, tap density, apparent density, and morphology may be measured and regularly monitored during continued use. ASTM F3049, "Standard Guide for Characterizing Properties of Metal Powders Used for Additive Manufacturing Processes," specifies further testing recommendations. Any additional powder characterization requirements should be as specified (see 6.2).

6.4 Definitions.

6.4.1 Build cycle. Definition of build cycle is in accordance with ISO/ASTM 52900, "Additive manufacturing — General principles — Fundamentals and vocabulary."

6.4.2 Powder lot. Definition of powder lot is in accordance with SAE AMS7002, "Process Requirements for Production of Metal Powder Feedstock for Use in Additive Manufacturing of Aerospace Parts."

6.4.3 Reuse powder. Definition of reuse powder is in accordance with SAE AMS7031, "Batch Processing Requirements for the Reuse of Used Powder in Additive Manufacturing of Aerospace Parts."

6.5 Subject term (key word) listing.

CuNi (70-30)
3-D Printing
Casting Substitute
Metal Powder Printing
Metal Printing
Powdered Metal

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH

Preparing activity:
Navy – SH
(Project AMPR-2026-003)

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